An outstanding feature of the carbon dioxide laser is its high efficiency: it is possible to convert electrical power to laser output power with an efficiency of greater than 10 percent.

The pertinent energy levels of carbon dioxide are shown in Figure 1-5. The details of the excitation mechanism are not completely understood at this

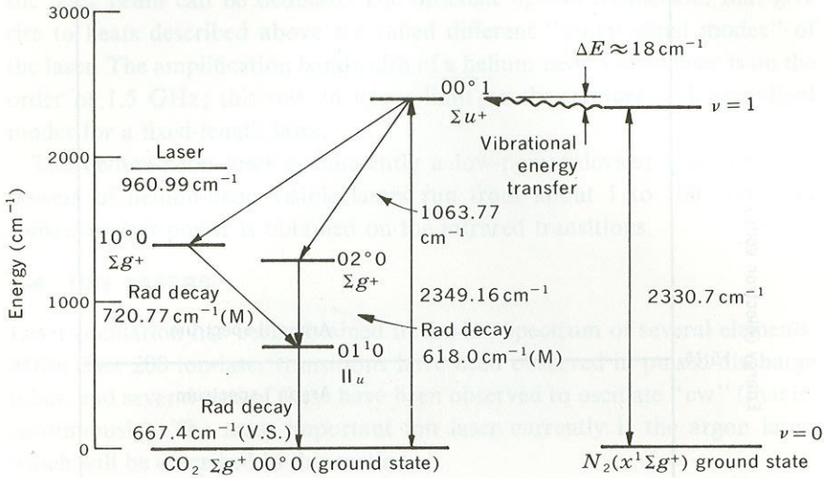


FIG. 1-5. Energy-level diagram for the CO_2 laser. [*From C. K. N. Patel, Phys. Rev. Letters,* **13**:617 (1964).]

time; apparently a major excitation process involves collisions of the first kind which directly populate the upper laser state. To obtain high output powers from carbon dioxide lasers, it is necessary to add other gases to the discharge. The two most common "additives" are nitrogen and helium. The enhancement obtained from the nitrogen is believed to be caused by population of the upper laser state by collisions of the second kind between metastable nitrogen molecules and ground-state carbon dioxide molecules. The effect of helium on the carbon dioxide laser is apparently to increase the decay rate of the lower laser state by collisional transfer of energy.

The characteristics of the output beam from a carbon dioxide laser are, in principle, similar to those of other gas lasers, but the high power of a carbon dioxide laser often gives rise to effects that are qualitatively different from those obtained with other gas lasers. For example, the usual result obtained when a carbon dioxide laser beam is scattered from a white card is that the card bursts into flame; the details of coherent scattering are of little interest in this case.

REFERENCES

1. W. R. Bennett, Jr., *Appl. Optics Suppl. on Optical Masers*, **1962**: 24; W. R. Bennett, Jr., *Appl. Optics Suppl. on Chem. Lasers*, **1965**: 3; A. L. Bloom, *Appl. Optics*, **5**: 1500 (1966).

These three papers are review articles which summarize progress in the general field of gas lasers. They are well written, and should be read by everyone interested in gas lasers.

2. W. B. Bridges and A. N. Chester, *IEEE J. Quantum Electronics*, **QE-1**: 66 (1965); E. I. Gordon, E. F. Labuda, R. C. Miller, and C. E. Webb, in P. L. Kelley (ed.), "Physics of Quantum Electronics," p. 664, McGraw-Hill Book Company, New York, 1966; E. F. Labuda, E. I. Gordon, and R. C. Miller, *IEEE J. Quantum Electronics*, **QE-1**: 273 (1965).

The papers by Gordon et al. discuss excitation mechanisms and operating characteristics of ion lasers. The article by Bridges and Chester describes the various ionic transitions on which laser action has been obtained.

3. C. K. N. Patel, *J. chim. Phys.*, **1967**: 82.

This is a recent summary of the state of CO_2 laser development. It is not comprehensive; much of the background work is described in the references cited at the end of the paper.

4. E. F. Labuda and E. I. Gordon, *J. Appl. Phys.*, **35**: 1647 (1964); A. D. White and E. I. Gordon, *Appl. Phys. Letters*, **3**: 197 (1963); E. I. Gordon and A. D. White, *Appl. Phys. Letters*, **3**: 199 (1963).

These three papers describe the detailed excitation mechanisms and scaling laws for the helium-neon visible laser, a subject which is not described in this book. They are important papers, although they have apparently been overlooked by many workers. Among other things, they show that the widespread belief that the output power of helium-neon lasers is limited by neon metastable states is erroneous.